Louis Mascari

Full Stack Software Engineer

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EXPERIENCE

Songtrust , *Full Stack Software Engineer · Downtown Music, part of Universal Music Group* 2024 – Present

- Building a greenfield multi-tenant GraphQL federation platform to integrate data across Downtown Music's operating companies (TypeScript, Node, Next.js, Apollo Federation).
- Own the frontend and ship features end to end across the API layer: the onboarding flow, tenant management, feature flags, access permission grants, and the integrations UI.
- Review code across the stack and make architectural and UI design decisions alongside engineering, Product & Design.
- Integrate third-party APIs that bring each operating company's data into a common interface, and built a guided workflow for data ingestion, AI-assisted transformation, and export.
- Set up the platform's engineering foundation: app shell and shared component library, internationalization, accessibility checks, pre-commit and pre-push hooks, and GitHub Actions CI/CD with deterministic merge gates.
- Co-develop team's AI development harness with head of technology: agentic workflows, orchestration & verification gates.
- On Songtrust's royalty administration platform, led delivery of the Business Signup Migration and Draft State epics with Product and Design, and migrated the core song submission flow from legacy Django templates to React.
- Cut the royalties dashboard's JS bundle 71% via code splitting; batched concurrent GraphQL queries into single requests.
- Profiled slow queries with EXPLAIN ANALYZE and fixed them with rewrites and indexes (songwriter search 13x faster).
- Hardened authentication: Twilio MFA, short-lived JWTs, idle timeouts & user-enumeration prevention.

Professional Case Management · Upstream Rehabilitation, *Physical Therapist* 2019 – 2023

- Managed complex caseloads and supervised assistants on interdisciplinary teams in underserved clinics and home health.

PROJECTS

MangoWave , *Creator* 2026 – Present

- Built a React audio visualizer on a fork of butterchurn (a WebGL 2 MilkDrop port) with 17 documented engine patches: GPU memory leak fixes, frame-rate normalization for 60–144Hz displays, and a fix for a 3–5 second mobile startup freeze.
- Compiled an HLSL-to-GLSL converter to WebAssembly and ran it in Web Workers, growing the library from 395 to 832 presets; integrated a WebAssembly GLSL optimizer that strips the ~96% of each converted shader that is dead code.
- Added cross-device rooms synced peer-to-peer over WebRTC, a serverless AWS backend (CDK, Lambda, DynamoDB) for Spotify OAuth and cloud sync, Playwright E2E tests in GitHub Actions, and localization in 9 languages.

Utilization Management Tool, *Creator* 2026 – Present

- Building an AI-assisted clinical review tool for healthcare: it reads a consult request, retrieves the matching clinical criteria, and drafts an approve or deny decision with references for a physician to finalize.
- Built the pipeline on AWS (FastAPI, Lambda, SQS, CDK): Textract OCR, RAG over a pgvector criteria vault, and Claude on Amazon Bedrock with Guardrails, with a Next.js reviewer console and facility portal behind Cognito role-based login.

MIMIC , *Co-creator & case study author* 2024

- Co-created, with a team of four, a self-hosted session replay framework: a Python CLI installer (PyPI) injects a capture script into an app's pages to record DOM changes, user interactions, and network requests.
- Sessions buffer in Redis, archive to S3 via pub/sub with PostgreSQL metadata & replay in a React dashboard.

SKILLS

Frontend: TypeScript, JavaScript, React, Next.js, Apollo Client, Tailwind CSS, MUI, HTML/CSS, WebGL, WebAssembly, WebRTC

Backend & Data: Node.js, Express, Python, Django, FastAPI, GraphQL (Apollo Federation), REST, SQL, PostgreSQL, Redis

AI & Agentic: Claude Code, Codex, Gemini, agent orchestration, LLM integration (Amazon Bedrock, Claude), RAG (pgvector, embeddings), Bedrock Guardrails

Cloud & Tooling: AWS (CDK, Lambda, API Gateway, ECS, S3, CloudFront, SQS, Textract), Docker, GitHub Actions, CircleCI, Playwright, Cypress, Vitest, Jest, pytest, Sentry, Datadog

EDUCATION

AWS Certified Solutions Architect – Associate, *Amazon Web Services* 2026

Bachelor of Science, Computer Science, *Western Governors University* 2025

Software Engineering & Full-Stack Web Development, *Launch School* 2022 – 2024

Doctor of Physical Therapy, *Emory & Henry University* 2016 – 2019